AUSTRALIAN SYSTEM SAFETY CONFERENCE 2026 · INDUSTRY PRESENTATION

The safety layer nobody shipped

Assuring accountability as agentic AI capabilities evolve



Payal Tiwari

Fractional Technology Advisor & Chief AI Officer

CROWNE PLAZA · ADELAIDE · OCTOBER 2026

ACS

- ACS is Australia’s largest and longest-serving tech community, representing professionals across industry, government and education. As the trusted leader of the tech sector, ACS champions the technologies, people and skills critical to Australia's future.
- The key aim is to grow the nation’s digital skills capabilities, industry capacity and inspire the next generation of technology professionals. The ACS Technical Committees are a vehicle by which this vision is supported.
- This presentation has been authored by Members of the ACS Technical Committees representing the personal views of the presenter(s) and should not be interpreted as representing the official position of the organisation.

Author Name: Payal Tiwari
Fractional Technology Advisor & Chief AI Officer
Member, ACS Sustainability Committee
BE (Computer Systems), Executive MBA
Postgraduate Diploma in Management
transformationtiwari@gmail.com
linkedin.com/in/payaltiwari

Contact ACS

www.acs.org.au

A NOTE BEFORE WE START

Part of today's talk describes stalking and family violence. If you need to step out at any point, that's fine.

WHAT YOU'RE ABOUT TO SEE

01

One incident report

02

One national failure

03

One question for agentic AI

01 — INCIDENT REVIEW

Every component performed as designed

Component	Designed for	Used for	Status
Location services	Navigation, convenience	Tracking where I lived	NOMINAL
Mobile network	Staying reachable	Relentless unwanted contact	NOMINAL
LinkedIn	Professional networking	Monitoring my career	NOMINAL
Instagram	Sharing with friends	Watching my stories and movements	NOMINAL
Mutual contacts	Social connection	Relaying information after I blocked him	NOMINAL

Every status nominal. The harm was not.

THE COST OF “WORKING AS DESIGNED”

What nominal looked like from the inside

5

relocations with my son

6

years to do it in

35+

calls in a single stretch

7

years of intervention orders

WHAT THIS REALLY IS

Not one incident. Infrastructure.

The same tools we use for work, connection and identity, turned into a surveillance system.

NOT AN EDGE CASE

What frontline practitioners see

99.3%

of domestic violence practitioners have clients experiencing technology-facilitated abuse

+245%

increase in practitioners reporting GPS tracking of victim-survivors since 2015

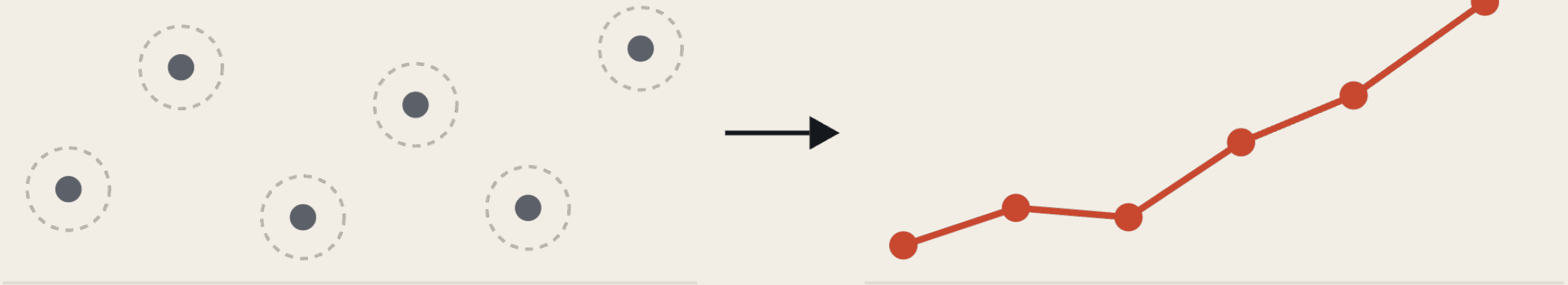
Rarely

enforced: intervention order breaches made through technology

Source: Wesnet & Curtin University, Second National Survey on Technology Abuse and Domestic Violence in Australia (2020), n = 442 practitioners

02 — A PATTERN-LEVEL HAZARD

The pattern is the hazard

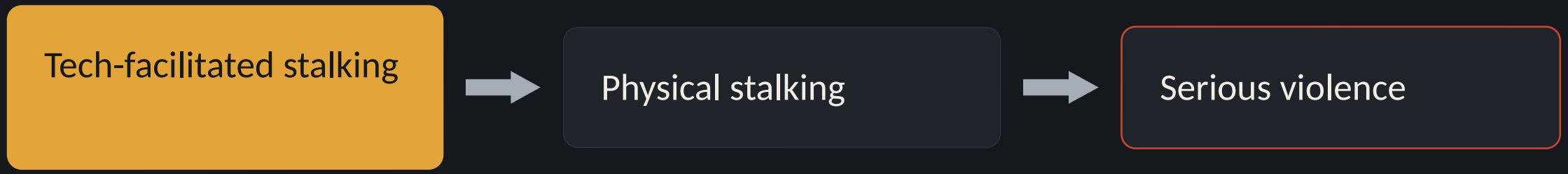


Assessed one at a time: minor, explainable, almost nothing.

Connected across years: calling, following, watching. Escalating. = Persistent Tech stalking

03 — THE LEADING INDICATOR

The signal risk assessment exists to catch



Technology-facilitated stalking is a well-documented predictor of escalation. Treating it as a lesser-tier concern filters out the leading indicator.

04 — DATA CONTINUITY

I kept my phone number

“Why don’t you just block him?”

“Why don’t you take a break from social media?”

What that advice costs

Isolation, which is exactly what power and control is looking for. And a broken evidence trail.

What continuity gave me

Every incident tied together. A pattern for police, not disconnected complaints.

05 — THE ASSURANCE BURDEN

Who has to prove the system should act?

The system waited for me to prove I was at risk.

Under MARAM, Victoria's family violence risk framework, priority depends on demonstrating risk consistently and in detail.

I gave statements so often that officers at my local station got to know me.

06 — WHO IS TELLING YOU THIS

Two careers, one story

THE TECHNOLOGIST

15+ years in technology transformation and strategy leadership roles

Deloitte (Director), EY (Senior Consultant), Accenture (Senior Consultant), Wipro Director, MUFG (Partner). Now a Fractional Technology Advisor and Chief AI Officer, and a member of the ACS Sustainability Committee.

LIVED EXPERIENCE

Living with technology-facilitated stalking

Intervention orders, relocations, and now advocacy in family violence policy and reform.

Those two things are not in tension. They are the same story.

07 — RISK CLASSIFICATION IN AN EMERGENCY RESPONSE

Three recognitions changed everything

01 · A JUDGE

Ruled it wasn’t safe for me to stay in my own home.

02 · HIGH-RISK DESIGNATION

1 hr+ → 8 min

Police response time once the risk was recognised.

03 · A SPECIALIST DETECTIVE

Secured through Specialist Violence advocacy, not my own.

That shouldn’t depend on luck, self-advocacy or postcode.

08 — YOU'VE SEEN THIS FAILURE BEFORE

Robodebt: my case, at national scale

$1.76bn
in debts raised

526,000
people affected

Unlawful
the Royal Commission's finding

	My case	Robodebt
System status	Nominal	Nominal
Burden of proof	On me	On citizens
Accountability arrived	When advocates stepped in	After a Royal Commission

Source: Royal Commission into the Robodebt Scheme, Report (2023)

09 — YOUR METHODS ALREADY NAME THIS

Hazards without a failure

STPA

Losses emerge from unsafe control and interactions

Not only from a component that breaks.

ISO 21448 · SOTIF

The intended function can be the hazard

Including reasonably foreseeable misuse, with no fault present.

AI6 · AUSTRALIA 2025

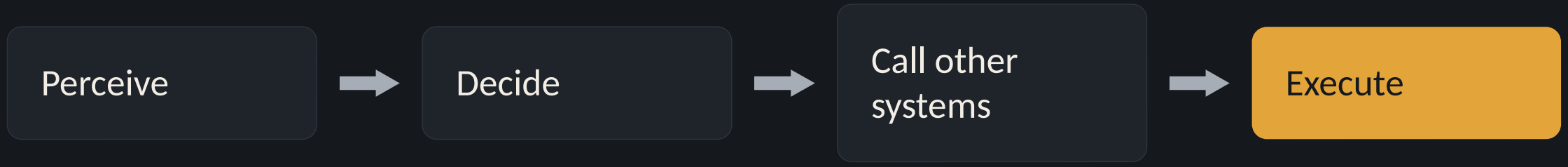
Make it easy to challenge or appeal outcomes

Practice 6 of the national Guidance for AI Adoption.

The concepts exist. They stop at the operator’s boundary.

Leveson, Engineering a Safer World (2011) · ISO 21448:2022 · National AI Centre, Guidance for AI Adoption (Oct 2025)

From recommending to acting



Agentic systems chain decisions, call other systems and act, often faster than any human can.

The question that matters: accountable to whom?

The layer missing from the stack

Accountability to the person the system acts on	LARGELY ABSENT
Enterprise value and adoption	WELL COVERED
System-level assurance and governance	WELL COVERED
Model performance and security	WELL COVERED

Based on 2026 agentic AI reports from PwC, McKinsey, IBM, Deloitte and BCG

12 — ALREADY IN YOUR SECTORS

Where the intimate adversary shows up

Transport

Connected-car apps that locate and unlock the vehicle.

Health

Shared records and portals that reveal an address or appointment.

Telecommunications

The account holder sees location, call logs and every line.

Energy and smart homes

Remote controls used to watch, disrupt and intimidate.

Emergency services

Risk triage that decides between 8 minutes and an hour.

Security

Threat models that assume the attacker is a stranger.

AI is a force multiplier for control

Stalkerware inside ordinary apps

Location sharing, weaponised

Fabricated “evidence”, erased trails

SCALE · PERSONALISATION · PREDICTION

The same properties cut both ways. If we don’t design against misuse deliberately, the capability serves whoever holds the controls.

MORE THAN AN ETHICS QUESTION

Accountability is a safety-critical property

So it belongs in hazard analysis, safety cases and assurance, not only in ethics statements.

Harm Distribution Review

1. Who carries the residual risk?
2. Can they see it?
3. Can they contest it?
4. Can they escape it?

A review step that asks not only how likely harm is, but **where it lands**.

15 — WORKED EXAMPLE (ILLUSTRATIVE)

An agentic assistant at a telco

A verified account holder asks the agent to “show where the family’s phones are” and “move the second line to my name”.

Review question	Agent’s view: nominal	What the review surfaces
Who carries the risk?	An authorised request	The person on the second line
Can they see it?	Task completed	No notice their location was shared
Can they contest it?	Standard complaints	Not while the requester controls the account
Can they escape it?	Account rules	Their number now belongs to someone else

Control: location and ownership changes on shared accounts pause for a safety check, and every line holder gets their own channel.

Four additions to the hazard log

H-01

Model the intimate adversary

They may know the passwords, hold the phone, and be a legitimate user of your platform.

H-02

Assure through life, not at release

Evolving systems need monitoring that hears from the people affected, not just operators.

H-03

Design for patterns and contestability

Connect small events. Give people a way to object that works in a moment of fear.

H-04

Consult the people who carry the harm

Victim-survivors are subject-matter experts in how your system fails.

IF YOU REMEMBER THREE THINGS

01 Recognise the pattern, not the incident.

02 Put accountability to the affected person inside the safety case.

03 Never let the person with the least power carry the burden of proof.

BACK TO THAT INCIDENT REPORT

Every status is still nominal.

Same number. Same platforms. What changed my outcome wasn’t the technology. It was people and systems that recognised the risk.

That layer can be designed. That’s the work.

THE QUESTION IN FRONT OF US

Not “does it work?” but “who does it work on, and who answers for it?”

Testing for use cases - (not the most common one) may help!

Thank you

Questions and conversation welcome



Payal Tiwari

Fractional Technology Advisor & Chief AI Officer
Member, ACS Sustainability Committee
BE (Computer Systems), Executive MBA
Postgraduate Diploma in Management
transformationtiwari@gmail.com
linkedin.com/in/payaltiwari

AVAILABLE FOR

AI governance and safety assurance advisory

Fractional CIO and CAIO engagements

Board advisory, keynotes and workshops

If today raised anything for you: 1800RESPECT · 1800 737 732 · available 24/7